Manuscript outline

Cell enlargement, polarity competition and migration persistence

Vision-aligned manuscript outline with proposed main figures and detailed legends

17 October 2026 · Revised after independent scientific and editorial assessment

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

Proposed argument: establish model- and stimulus-dependent size effects in finite polarity replacement; then connect specific contributions to local input response, incumbent restoration, transport, mechanics and site competition. Existing evolved interventions distinguish some causal contributions; the common gain/recovery assays provide comparable, mechanism-linked observables rather than a causal partition of all three duration curves. Supported rings test placement and sustained competition. Replicated noisy rings measure activity memory, with physical displacement, magnitude and observation coverage alongside angular descriptors. Published migration observations and conditional OU velocity-memory estimates supply separate empirical motivation. The new WGD motility assay remains a major pending contribution.

This is a proposed manuscript outline, not a completed manuscript. The order of the mechanism and existing-migration components remains an editorial choice. Six main figures are numbered for navigation, with continued sheets where needed for legibility. The second edition adds an all-five mechanism-to-measurement crosswalk, physical-motion and adaptation diagnostics, and the requested migration curve-shape display. Mechanism-linked cellular experiments are proposed, and Figure 6 remains wholly unmeasured.

Suggested introduction: distinguish migration persistence from speed; describe why WGD-associated enlargement motivates a geometric polarity question; introduce established biochemical and mechanochemical models as complementary hypotheses; explain the need to connect mechanisms to measurable activity/protrusion behaviour

Manuscript outline · continued

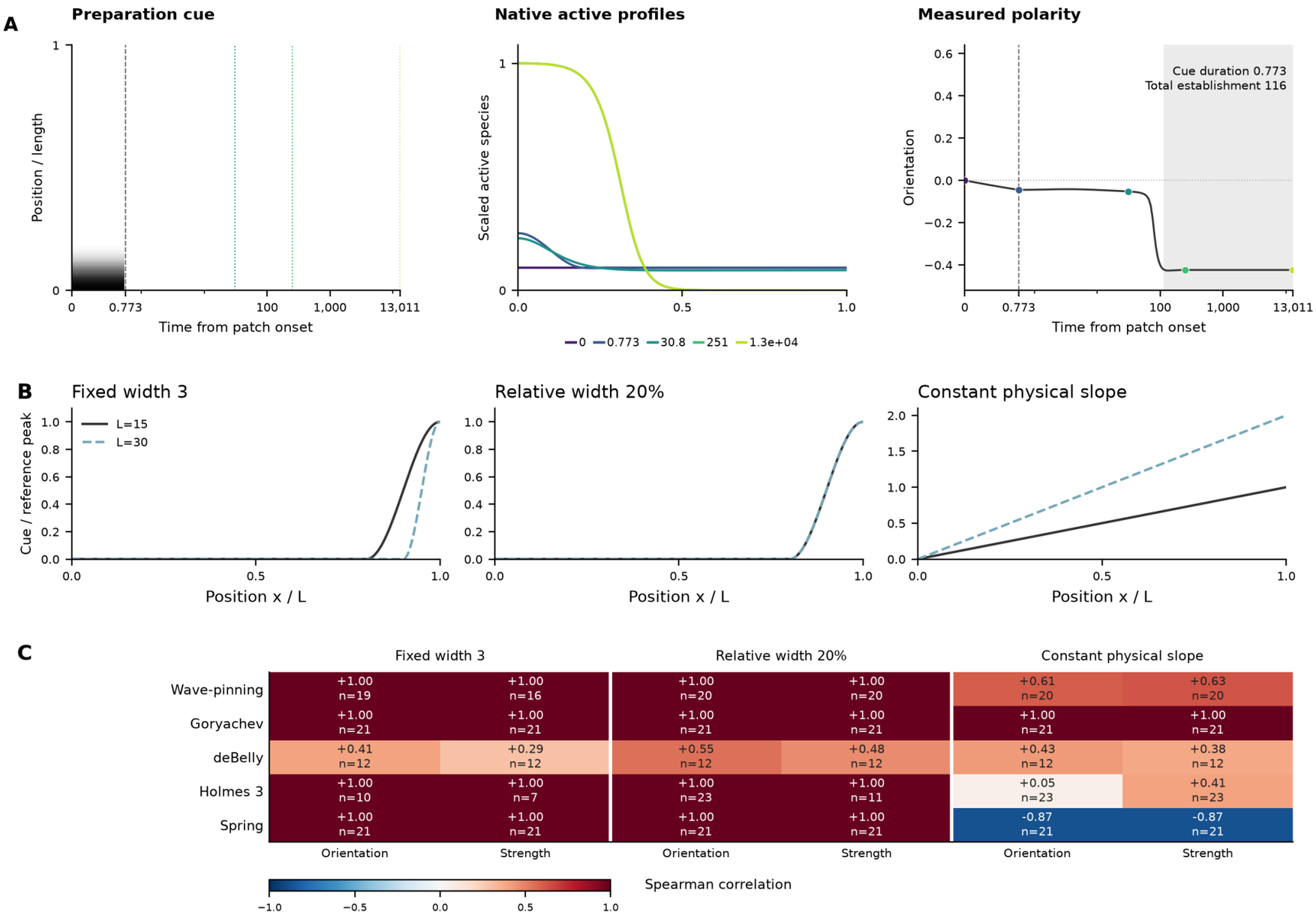
without assuming a cell-velocity law.

Suggested discussion: retain nonmonotone, null and opposite responses as informative model differences; separate deterministic cue resistance, noisy activity memory and velocity memory; identify concrete experiments; discuss finite horizons, stimulus geometry, conservation conventions and native times. The conclusions should remain conditional until the new motility and mechanism-validation experiments are obtained.

Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Incumbent organization, local response and model-specific mechanisms of competition
- Figure 3. Challenger placement changes sustained competition and global activity polarity on supported rings
- Figure 4. Activity memory, physical site motion and forced-state qualification on noisy rings
- Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation
- Figure 6. RESERVED: new WGD motility-assay results

Figure 1. Five models and three finite challenges



Current canonical comparison, 10 October Spearman revision; corrected independent-endpoint deBelly.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Example wave-pinning preparation from a homogeneous state using a finite left patch. The cue ends at 0.773 native time units, whereas the subsequent basal relaxation establishes the polarized profile later (reported establishment arrival about 116). Coloured profiles and orientation samples distinguish input duration from autonomous development. The grey late interval is verification, not additional stimulation. Profiles are scaled for display.

B, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

C, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

Sources and reproducibility

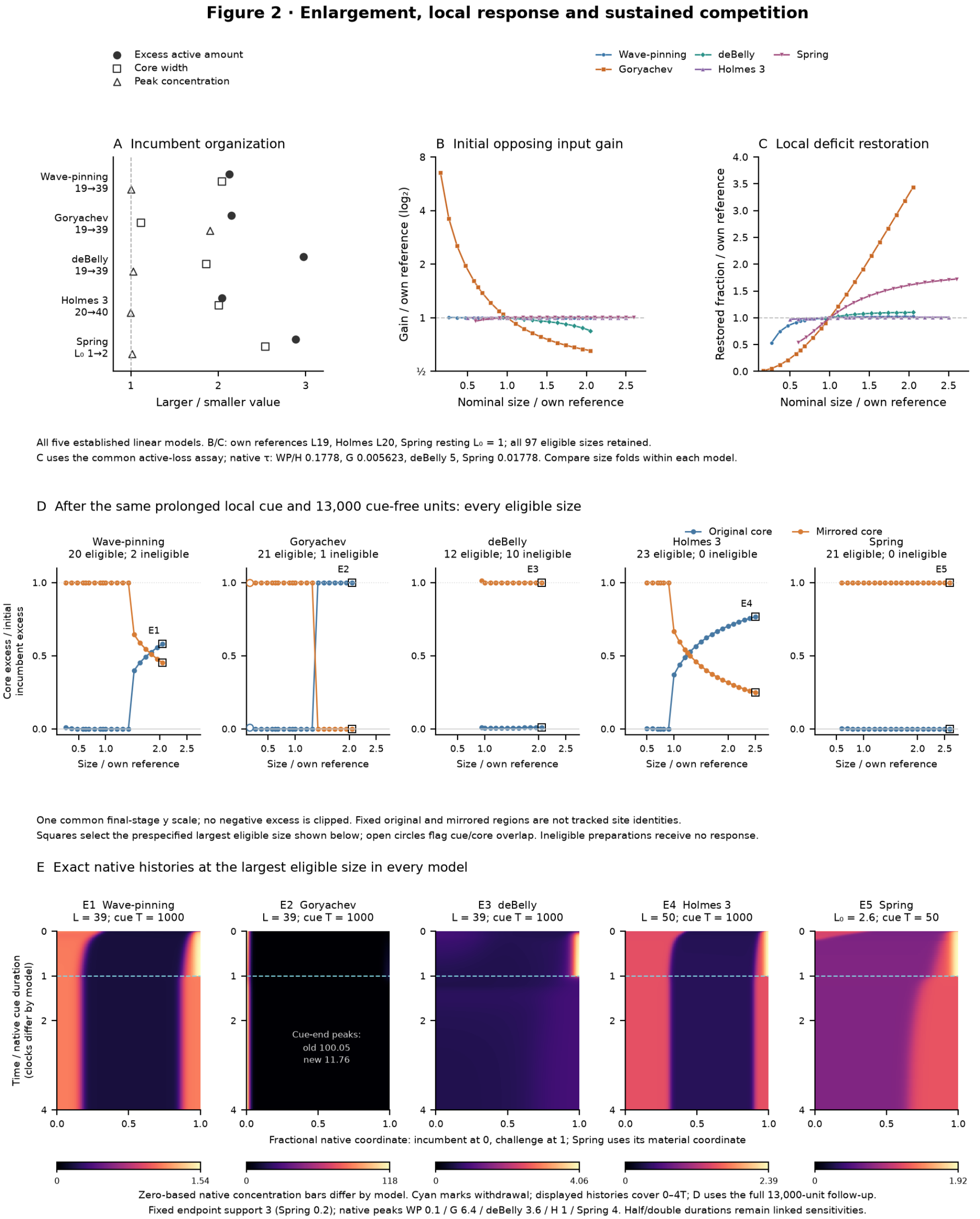
Three-test protocol and Spearman rule

Correlation table

Corrected endpoint method

Current duration curves

Figure 2. Linear response and mechanistic evidence · image 1 of 3



All-five linear results, with the title renumbered for this outline; scientific pixels and numerical inputs are preserved.

Figure 2 continued: mechanisms, evidence and measurements · image 2 of 3

FIGURE 2 CONTINUED · EXISTING MECHANISTIC EVIDENCE · SHEET 1/2

Mechanistic explanations and their corresponding measurements

Companion to manuscript Figures 2–3. Existing mechanistic evidence explains specific contributions and motivates measurements; the common gain/recovery panels do not causally partition all three cue-duration curves.

Model / candidate mechanism	Observed evidence	Corresponding measurements
Wave-pinning A wider incumbent and conserved-pool competition make challenger growth and post-cue survival depend on spatial reorganization, even when initial local activation gain changes little.	At L15, width-3 pulses of 20/60 native units collapse/retain the challenger. At L39, fourfold inactive diffusivity rescues the native 5120-unit fixed-support failure. Gain and local deficit repair remain near-null with size.	Measure incumbent extent and excess signal, early challenger growth, boundary retreat and both sites after withdrawal. Pair local activity with substrate-recovery or redistribution measurements; distinguish peak height from region expansion.
Goryachev A taller incumbent sequesters conserved material, reducing inactive concentration at the challenger while increasing the incumbent's measured response to specified damage. Initiation and incumbent restoration are coupled contributions.	For L19→39, fixed-support initial gain falls to 0.654 and common active-loss repair rises 3.439-fold. Separate L9→39 inactive-bolus removal/return gives a 15.71-fold increase in immediate reaction per deficit.	Measure incumbent peak, physical extent and integrated signal; measure challenger activation under characterized input and available substrate. Use a paired small activity-loss/sham assay to distinguish immediate local restoration from subsequent recovery.

Inference scope

Wave-pinning: Withdrawal examples and an evolved fixed-dose mobility intervention support transport/domain competition. They do not prove constant front speed or quantify a new duration threshold.

Goryachev: Native input multiplies inactive concentration; equal cue peaks therefore differ in biochemical activation. Repair is perturbation- and clock-specific. Neither contribution supplies an exhaustive or additive allocation of reversal duration.

Source: mechanism_crosswalk_v2.json · Exact assays, gates and source citations accompany the full legend.

Figure 2 continued, mechanism-to-measurement companion, sheet 1 of 2. Wave-pinning, Goryachev. Existing evidence and inference scope remain distinct.

Figure 2 continued: mechanisms, evidence and measurements · image 3 of 3

FIGURE 2 CONTINUED · EXISTING MECHANISTIC EVIDENCE · SHEET 2/2

Mechanistic explanations and their corresponding measurements

Companion to manuscript Figures 2–3. Existing mechanistic evidence explains specific contributions and motivates measurements; the common gain/recovery panels do not causally partition all three cue-duration curves.

Model / candidate mechanism	Observed evidence	Corresponding measurements
Holmes 3 Coupled Rac/Rho domains and cue coverage permit a locally activated challenger to coexist with an incumbent. Broadening and domain competition remain important even when local gain and damage repair change little.	Fourfold inactive mobility fails to rescue fixed-support L28/L50 at 5120 units; relative-20% cues pass orientation at all eligible sizes. Ring placement separates merger from distinct sites within one complete C40 basal state; a C30 example has a cue-associated survivor.	Image both Rac-like and antagonistic activity regions; vary input position and support at fixed measured size. Track site extent, separation, merger and extinction through withdrawal alongside the global reporter vector and its magnitude.
Corrected deBelly Length-dependent resting mechanics and old-end MCA depletion delay incumbent suppression; at the longest lengths, slower local MCA accumulation adds a further delay. MCA-to-Rac inhibition connects this local organization to activity.	Relative-20% cues: L23→31 old-end MCA starts 0.05348→0.04823; by 180 s it reaches 0.06451→0.06044. The 10% Rac-decline clock is 123.49→142.97 s, versus 637.39 s at L39. Corrected resting tension-to-Rho ablation reduces displacement spread 50.4%.	Measure old/new-end activity, attachment-reporter level, local flow and protrusion together. Compare basal depletion and absolute common-time refill with incumbent decline; quantify local transport/replacement balance. Perturb tension coupling separately to test the starting-state mechanism.
Spring Extension-dependent global deactivation couples mechanics to polarity competition. A constant-slope cue also grows in peak with size, helping explain why its length trend differs from the increasing local-patch requirements.	Setting the extension-dependent deactivation coefficient to zero only during stimulation defeats 4/4 native successful gradient pulses. Matching the ramp peak defeats 3/4. Under fixed/relative patches, two longer sampled cells still reverse after the same deactivation-term removal.	Measure physical extension, global activity and local incumbent/challenger histories during the same characterized cue. Compare constant local support with increasing spatial coverage; test a specified mechanical or activity-feedback perturbation while recording its actual effect on shape.

Inference scope

Holmes 3: Fixed-dose controls and matched-state placement support spatially structured competition; mobility is not a universal rescue. Relative support changes exposure. Orientation success, strength restoration and finite local fate are different readouts.

Corrected deBelly: Endpoint budgets explain coupled trajectories; the separate evolved ablation establishes a contribution to resting mechanics. Neither proves a universal MCA withdrawal threshold or a causal share of reversal duration. Low-length decreases and L39 require separate treatment.

Spring: Evolved controls establish geometry-dependent contributions at fixed native doses, not counterfactual duration thresholds. Endpoint motion, geometric dilution and native recovery remain; this is not removal of all mechanics.

Source: mechanism_crosswalk_v2.json · Exact assays, gates and source citations accompany the full legend.

Figure 2 continued, mechanism-to-measurement companion, sheet 2 of 2. Holmes 3, Corrected deBelly, Spring. Existing evidence and inference scope remain distinct.

Figure 2. Incumbent organization, local response and model-specific mechanisms of competition

Mechanistic explanation joined to specific activity measurements: organization, input gain, conditional local recovery and actual finite competition in the established linear framework.

Detailed legend

A, Larger/smaller ratios of incumbent excess-active amount, half-contrast core width and absolute peak concentration. Pairs are L19→39, Holmes L20→40 and Spring resting L_o1→2. Amount is the above-background integral in the incumbent core; it is not an exact width × peak product. Goryachev mainly raises concentration, while the other models predominantly broaden the incumbent. deBelly uses corrected independent endpoints; Spring uses its actual physical domain.

B, Cue-weighted incremental primary-activity production per unit native opposing input, shown for all 97 qualifying basal checkpoints. Native gain is inactive substrate for Wave-pinning/Goryachev/Spring, inactive Rac divided by 7.5 for Holmes and $0.04/(1+Rho^2)$ for deBelly, which has no explicit inactive Rac pool. Gain and nominal size are divided by each model's reference: L19, Holmes L20 or Spring L_o1. The shared logarithmic gain axis preserves nearly unchanged responses.

C, Sham-relative deficit restoration after the same small active-deactivation perturbation at every qualified size. Remove 0.5% of a smooth compact taper of the frozen basal incumbent excess profile; 0.1%/1% amplitudes check linearity. Transfer removed material pointwise to the explicit inactive pool where present; deBelly changes Rac alone, initially preserving MCA/mechanical/controller states. All native fields then evolve without an opposing cue. Integrate the deficit between sham and probe in the fixed original core and measure its repaired fraction at one fixed clock per model. Plot that fraction relative to the same reference as B. Native clocks are 0.1778279410 for Wave-pinning/Holmes, 0.00562341325 for Goryachev, 5 for deBelly and 0.01778279410 for Spring. These clocks do not calibrate biochemical rates across models.

D, Every qualifying size after a fixed opposing cue and 13,000 cue-free units. Endpoint raised-cosine support is 3 (Spring 0.2), peaks are 0.1/6.4/3.6/1/4 for Wave-pinning/Goryachev/deBelly/Holmes/Spring, and duration is 1000 native units (Spring 50), unchanged across sizes. Original and mirrored core excess amounts use the frozen basal background and divide by initial incumbent excess. One final-stage y scale retains signed values; negative excess means activity below basal background, not negative concentration. Open circles flag cue/core overlap. Coverage is 97 qualified states and 13 preparation-ineligible sizes; no response is invented for an ineligible start.

E1–E5, Exact native histories at the prespecified largest qualifying size of each model: L39/L39/L39/L50/Spring $L_o2.6$. Squares in D identify these choices. Horizontal coordinate is fractional native position, with Spring in material coordinates; time is divided by each native cue duration. Cyan marks withdrawal. The displayed history covers 0–4 cue durations, whereas D uses the complete 13,000-unit follow-up. Zero-based activity color

Figure 2. Incumbent organization, local response and model-specific mechanisms of competition · continued

scales are model-specific. Fixed-support sequences do not replace the previously specified relative-20% deBelly illustrations.

All 97 original full-span recovery archives qualify at their fixed clocks, with numerical sensitivity, amplitude-linearity and unsaturated-prefix checks. The separate Spring early-window amendment is additional validation and is unused by the main collection. The finite campaign contains 97 primary branches plus 30 prespecified half/double-duration anchors. The new histories reproduce all original 10,000-unit orientation, scalar-strength and field-tail flags; those are distinct finite-window diagnostics.

Continued mechanism-to-measurement sheet. Companion to manuscript Figures 2–3. Existing mechanistic evidence explains specific contributions and motivates measurements; the common gain/recovery panels do not causally partition all three cue-duration curves.

This companion joins mechanisms to corresponding measurements without treating the common active-loss assay as a complete explanation of all duration curves. An evolved intervention, a fixed-state reaction/source calculation, a local budget and a descriptive temporal association provide different evidence. Native units, perturbation directions and clocks remain model-specific. Proposed cellular measurements correspond to these mechanisms but would not uniquely confirm a model.

Wave-pinning: the L15 illustration uses a fixed-support-3 right raised-cosine patch at native peak 0.1, from the exact corrected nominal relative-20% basal checkpoint. The 20-unit pulse collapses after withdrawal; the 60-unit pulse produces a retained reversed state. These bracket the sampled 35.5-unit strength threshold. In the separate original three-test fixed-support control at L39, the same 5120-unit pulse fails natively and passes after fourfold inactive diffusivity throughout pulse and recovery; a 401-cell remesh alone remains a failure. The scalar orientation gate uses the final 500 of 10000 cue-free native units: orientation above 0.05, contrast at least 0.05 and orientation range below 0.005. Strength additionally restores the initial scalar magnitude to within 1%; the native-field flag remains separate. A fixed-dose rescue does not locate an intervention-specific threshold.

Goryachev: current main Figure 2 measures gain with fixed support 3 and peak 6.4 and common incumbent active-loss recovery at tau=0.00562341325. The L19-to-L39 gain/repaired-fraction folds are 0.653611 and 3.43942. The older whole-inactive-profile experiment is a separate perturbation: remove 0.5% of the complete frozen inactive profile, retain native kinetics for 0.1 native units, then return the removed profile exactly pointwise. In the frozen original half-height active core, the sham-relative immediate net-reaction response per active deficit is 0.9730925 at L9 and 15.2853874 at L39, a 15.70805-fold ratio. Its 90%-damage-recovery clocks are 18.8717 and 2.6400; do not attach the slower late-tail statement from the different activation-suppression assay to

Figure 2. Incumbent organization, local response and model-specific mechanisms of competition · continued

this primary bolus experiment. These coupled substrate/incumbent effects are not independent fractions of reversal duration.

Holmes: fixed-support-3 countercues have native peak 1. Fourfold inactive diffusivity changes both Rac and Rho inactive pools during pulse and recovery. At L28 and L50 the 5120-unit cap remains outside the scalar orientation gate; this control does not establish that no longer intervention pulse could reverse. Relative-20% responses use a larger physical cue support as length increases and pass orientation at all eligible lengths, while original-strength replacement can remain unavailable. Ring C40 d12 and d16 histories begin from the identical complete basal Rac/Rho state, receive the same full-support-3, peak-1, 1000-unit input and are read through withdrawal. Placement changes both boundary routes and cue angle; no pure-distance causal interpretation is implied.

Corrected deBelly: all endpoint values are concentrations in the first finite-volume cell, x=L/402, at the original incumbent; the challenged end is the last cell, x=L−L/402. Every trace starts from that length's exact corrected nominal relative-20% basal checkpoint, with independent local endpoint commands. The opposing right raised-cosine patch covers 20% of nominal length, peak 3.6, followed by 10000 native seconds of basal recovery. Selected shortest tested 1%-scalar-strength-passing durations at L23/L31/L37/L39 are 232.24490/285.28061/355.99490/809.08163 seconds, with separate native-field tail flags false/true/true/false. These are different experiments from the fixed-support 1000-unit common competition panel, whose later field flags must not be transferred here.

The corrected deBelly common-time comparison uses 180 seconds after opposing cue onset, before withdrawal at all four cited lengths. Old-end bound MCA starts at 0.05348320/0.04823201/0.04588120/0.04665557 and reaches 0.06450752/0.06044116/0.05351708/0.05066045 at L23/L31/L37/L39. Thus the 180-second increments are 0.01102432/0.01220915/0.00763588/0.00400488. The descriptive first 10% decline from each cell's initial old-end Rac occurs at 123.49009/142.97494/211.59354/637.39210 seconds; these are within-cue local clocks, not thresholds for withdrawal success. L39 begins less depleted than L37, so initial depletion alone cannot explain its delay. The relative-cue duration decreases from L18 to L21 and from L31 to L33; the incumbent-decline clock does not explain that complete nonmonotone curve.

The separate corrected deBelly resting ablation removes only the tension-dependent increment to Rho activation globally, from identical corrected basal checkpoints at L23/L31/L39 without a countercue. Native MCA-to-Rac feedback, Rac/Rho mutual inhibition, direct tension inhibition of protrusion, Rho active stress and independent endpoint relaxation remain. The displacement spread decreases from 1.968526 to 0.976774 micrometres after qualified settling and a 10000-second holdout. Rac contrast remains 0.2051–0.2103, although the L31/L39 normalized orientations lie below 0.05.

Figure 2. Incumbent organization, local response and model-specific mechanisms of competition · continued

This evolved intervention supports a contribution to resting mechanics, not a measured reversal-duration contribution. Endpoint MCA budgets separately retain profile advection, flow-gradient dilution, diffusion, net binding and uniform replacement; total-pool replacement includes the sequential exchange residual. Fixed-state mechanical source superposition is not an evolved pathway ablation.

Spring: native times/lengths have no biochemical or micrometre calibration. At resting lengths 0.6/1.6/2.2/2.6, the native successful fixed-slope pulse durations are 0.19/0.1275/0.125/0.12875. Setting only d1=0 during those pulses defeats all four orientation responses after native 10000-unit recovery. Matching ramp peak to the resting-length-1 reference defeats the latter three; the shortest still succeeds. Under fixed-support-0.2 pulses at lengths 0.6/1.6/2.6, durations 0.152/1.275/5.816320785, d1 removal defeats only the shortest. Under relative-20% pulses, durations 0.22375/0.615625/1.21875, it likewise defeats only the shortest. Moving-boundary motion and geometric dilution remain in all these controls. Native feedback is restored for recovery, and the same scalar orientation gate as the other original three-test controls is applied.

The ring and noisy-ring measurements extend these mechanism-linked readouts to site placement, local patterns and whole-boundary activity-vector magnitude/correlation. Angular memory, physical single-site displacement, finite deterministic cue resistance and conditional OU velocity memory remain distinct. The planned SUM-159 WGD+/− motility result is a separate pending empirical contribution.

Narrative points and conclusions

- All five models acquire more incumbent material, while its organization and functional consequences differ. Near-doubling gain folds are 0.65361 for Goryachev and 0.84570 for corrected deBelly; the other three are approximately one.
- The corresponding common-recovery folds are 1.02562 for Wave-pinning, 3.43942 for Goryachev, 1.10175 for deBelly, 1.00871 for Holmes and 1.60800 for Spring. These are within-model size effects along a declared active-loss disturbance, rather than cross-model stability rankings.
- Finite stimulation separates activation from later survival. At the longest intervals Wave-pinning and Holmes retain two threshold-defined regions, whereas Goryachev restores the original incumbent after transient challenger activity. All retained deBelly and Spring primary branches end right-oriented.
- deBelly exposes why cue-on amplitude and orientation alone are insufficient: L18 mirrored-core excess falls from about 7.99 initial-incumbent units during the cue to about 1.013 afterward. At 3000 cue-free units, 12/12 branches pass orientation, 5/12 scalar strength and 0/12 the field-tail flag, compared with 12/11/12 at 8000–13,000.

Figure 2. Incumbent organization, local response and model-specific mechanisms of competition · continued

- Mechanistic evidence is model-specific. Wave-pinning has an evolved fixed-dose inactive-mobility rescue; Holmes does not share that rescue at the tested cap. Spring has a cue-only extension-dependent deactivation intervention whose effect depends on stimulus geometry. These distinguish transport, spatial competition and mechanical feedback contributions without defining new counterfactual thresholds.
- Corrected deBelly has nonmonotone duration curves, including decreases at L18–21 and L31–33. Within selected rising ranges, endpoint MCA depletion and common-time accumulation help explain delayed incumbent Rac decline. L39 requires slower refill in addition to its starting state. A separate resting-state tension-to-Rho ablation reduces displacement spread by 50.4%; that causal result concerns resting organization, not an assigned fraction of reversal duration.
- Goryachev links lower available inactive concentration to weaker challenger activation and stronger incumbent repair under specified perturbations. The common active-loss assay, older inactive-bolus experiment and activation-suppression assay have different clocks and recovery tails and remain separate.

Proposed measurements and experimental follow-up

Measure incumbent physical extent, peak and excess integrated reporter signal against measured cell size. Molecular-amount interpretation requires calibration; spatial broadening and increased concentration-like signal should remain separate.

Apply a characterized local input and measure early challenger activation separately from its later establishment. Use a specified incumbent-targeted activity reduction and paired sham to measure actual local loss/rebound and shrinking sham-relative deficit over a declared window.

Track both sites during substantial fixed stimulation and after withdrawal, retaining extinction, return, merger, replacement and coexistence histories. Record local protrusion independently to test activity–protrusion correspondence; these experiments are proposed.

Wave-pinning: Measure incumbent extent and excess signal, early challenger growth, boundary retreat and both sites after withdrawal. Pair local activity with substrate-recovery or redistribution measurements; distinguish peak height from region expansion.

Goryachev: Measure incumbent peak, physical extent and integrated signal; measure challenger activation under characterized input and available substrate. Use a paired small activity-loss/sham assay to distinguish immediate local restoration from subsequent recovery.

Figure 2. Incumbent organization, local response and model-specific mechanisms of competition · continued

Holmes 3: Image both Rac-like and antagonistic activity regions; vary input position and support at fixed measured size. Track site extent, separation, merger and extinction through withdrawal alongside the global reporter vector and its magnitude.

Corrected deBelly: Measure old/new-end activity, attachment-reporter level, local flow and protrusion together. Compare basal depletion and absolute common-time refill with incumbent decline; quantify local transport/replacement balance. Perturb tension coupling separately to test the starting-state mechanism.

Spring: Measure physical extension, global activity and local incumbent/challenger histories during the same characterized cue. Compare constant local support with increasing spatial coverage; test a specified mechanical or activity-feedback perturbation while recording its actual effect on shape.

Scope, limitations and pending validation

A positive regional recovery coefficient does not establish generic stability or resistance to an actual challenger; deBelly's Rac-only kick leaves slower organization initially intact. Fixed cores do not uniquely track site identity.

Native clocks, length scales and input channels are not biological dose/time calibrations. Goryachev L39's core contains only four native cell centers; solver checks do not establish spatial convergence.

Half/double-duration controls establish bounded sensitivity, not a universal commitment threshold. Field-tail flags and finite retained patterns do not prove attractors or a cell-velocity law.

The crosswalk retains evolved interventions, fixed-state derivatives and temporal/budget associations at their respective inference levels. It does not assign additive causal fractions to full duration curves or identify a unique biological model from reporter measurements.

Model definitions and readouts

$$D(t) = A_{\mathrm{sham},\,\Omega_0}(t) - A_{\mathrm{probe},\,\Omega_0}(t) \tag{1}$$

The amount is integrated over the frozen original incumbent core; Spring uses each trajectory's physical length with fixed material-coordinate membership.

$$F_\tau = \frac{D(0) - D(\tau)}{D(0)}, \qquad K_\tau = \frac{F_\tau}{\tau} \tag{2}$$

Figure 2. Incumbent organization, local response and model-specific mechanisms of competition · continued

The figure shows restored fraction relative to its model-specific reference. A positive fraction means that one regional perturbation-induced deficit shrank relative to sham.

Sources and reproducibility

Frozen initial checkpoint intake
All-five actual results and measurements
Complete normalized linear gain
Complete qualified linear recovery
Finite competition and duration checks
Compact exact native display histories
Assembly provenance
Mechanism, evidence and corresponding measurement crosswalk
Native and mobility-control outcomes
Audited withdrawal examples
Separate inactive-bolus experiment
Bolus protocol and audit lineage
Matched-basal ring placement and finite fates
Corrected endpoint states and field flags
Descriptive incumbent-decline clocks
Endpoint mechanism and native budget lineage
Corrected resting ablation results
Independent resting ablation validation
Independent endpoint correction and exclusions
Exact intervention definitions
Native extension/deactivation equations

Figure 3. Supported-ring competition · image 1 of 2

Figure 3, continued sheet 1: matched native organization, input gain and conditional response. Only the working title numeral is changed from the delivered artwork.

Figure 3. Supported-ring competition · image 2 of 2

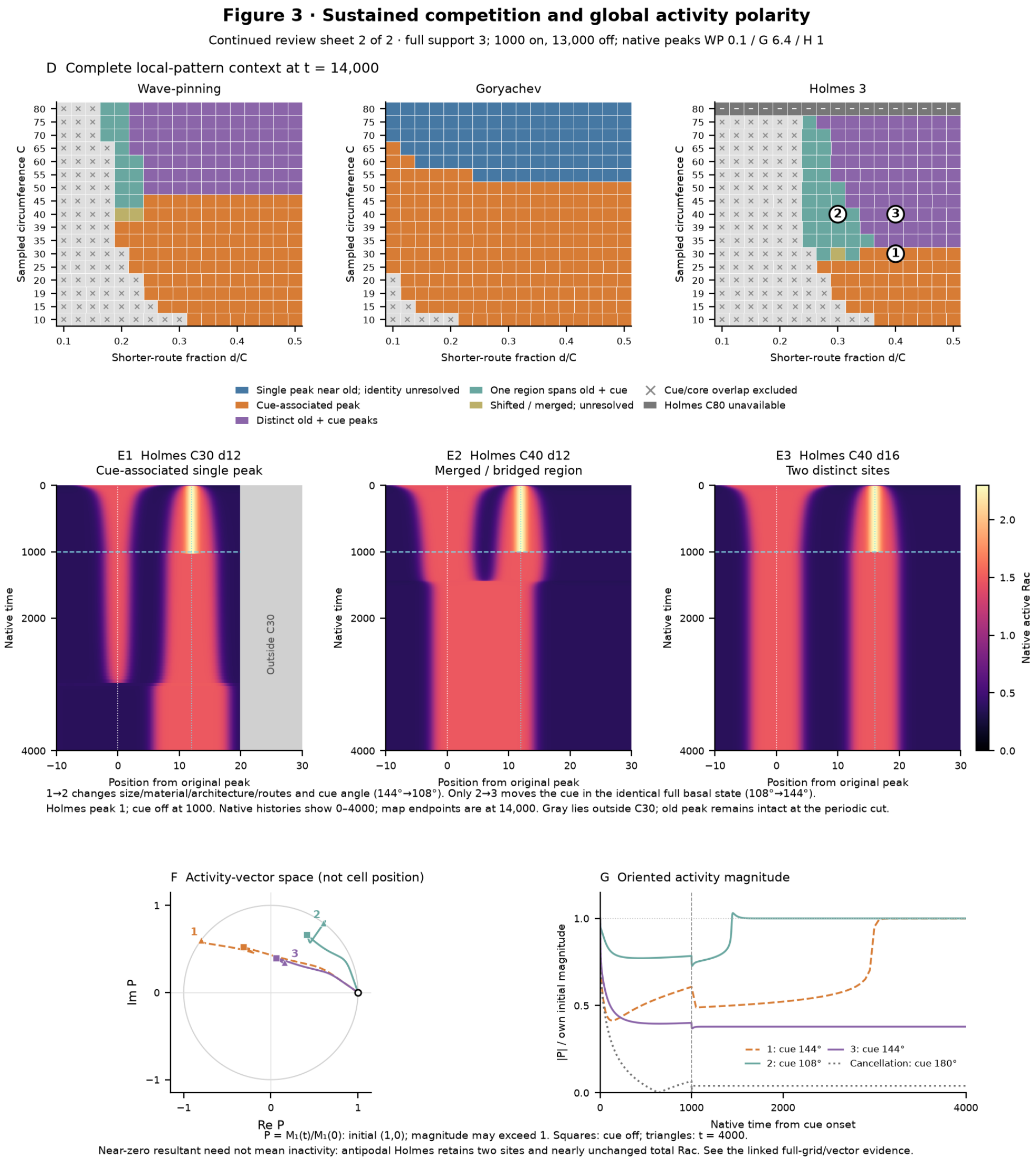


Figure 3, continued sheet 2: complete finite-pattern maps, linked histories and activity vectors. Preserved as one proposed manuscript figure across two review sheets. Only the working title numeral is changed from the delivered artwork.

Figure 3. Challenger placement changes sustained competition and global activity polarity on supported rings

Extend the established chemical models to a declared closed geometry, check interval/ring correspondence and connect local responses and site histories to measurable global activity asymmetry.

Detailed legend

A, Actual independently prepared interval L15 and ring C30 primary-activity profiles in Wave-pinning, Goryachev and Holmes. The interval is reflected for display. Across all 13 matched C=2L pairs, ring core width and excess-active amount are about twice the interval values; peak, matched gain and common recovery correspond closely. Maximum native recovery discrepancies are 0.04767%, 1.01636% and 0.000225% respectively. These are different native meshes and independent preparations, rather than identical reflected states or a continuum-convergence result. Matching endpoint support 3 requires full ring support 6; the finite ring challenges below use full support 3.

B, Initial local gain versus circumference C and shorter-route fraction d/C, divided by each model's C30 antipodal gain on one shared 0–3 scale. Rows are discrete sampled circumferences. C, The same active-deactivation assay as Figure 2 applied at exactly 10 native units after cue onset, with the same cue continuing in sham and probe. Divide the repaired fraction by that circumference's basal value, measured once per size. Frozen original core, perturbation direction, native clocks and complete coupled evolution remain unchanged. The ratio map does not remove the basal size trend; the linked raw basal curves retain it.

All maps preserve 867 planned cells: 652 complete challenges, 198 cue/core overlap exclusions shown by crosses and 17 unavailable Holmes C80 preparations shown in dark gray. The numerical recovery audit covers 50 basal and 652 challenged states and all 7020 branches. Eight early states lack a distinct old-associated site and their measurements concern the original region rather than a uniquely resolved incumbent.

D, Complete local-pattern geography after a full-support-3 raised-cosine cue for 1000 native units and 13,000 cue-free units. Peaks remain 0.1/6.4/1 for Wave-pinning/Goryachev/Holmes. Classes distinguish a sole old-associated maximum with unresolved identity, cue-associated peak, distinct old/cue peaks, one spanning region and shifted/unresolved patterns. Endpoint geography alone does not establish a site's lineage or an asymptotic branch.

The three numbered Holmes histories correspond exactly to displayed map cells: 1, C30 d12 develops a cue-associated single survivor; 2, C40 d12 forms a merged/bridged region; 3, C40 d16 retains distinct sites. They share a raw Rac scale and linear time from onset to 4000; cyan marks withdrawal at 1000. The periodic display keeps the original site at zero intact; gray padding lies outside C30. Only 2→3 changes cue placement in the identical complete basal state. Comparison 1→2 jointly changes material, architecture, both routes and cue angle (144°→108°); 2→3 changes cue angle from 108°

Figure 3. Challenger placement changes sustained competition and global activity polarity on supported rings · continued

to 144°.

The lower plots use the exact same histories to show the normalized complex activity moment in activity-vector space and its magnitude versus time. Initial vector is (1,0), squares mark withdrawal and triangles mark t=4000. Magnitude may exceed one. The antipodal Holmes safeguard has two active sites and nearly unchanged total active Rac despite a near-zero first moment. Direction is unavailable below relative magnitude 0.1; 0.05/0.2 sensitivity gates, total activity and first/second harmonic ratios remain linked. A second harmonic is not a unique site count.

Nine selected half/double-duration anchors keep their local endpoint class, but vector magnitude and angle can change. Full-grid readouts retain 18 class changes from 3000 to 13,000 cue-free units and four changes from 8000 to 13,000. Thus the main maps are declared finite observation slices, not a universal settling or duration-independent phase diagram.

Narrative points and conclusions

- Initial local gain varies by less than 0.09% across qualified placements at fixed circumference, while sustained competition produces sharply different local patterns. Input gain alone therefore does not determine later site fate.
- Every challenged conditional repaired fraction is positive even though the actual unperturbed original core loses active material during the same window. Challenged/basal ratios span 0.469–1.000 in Wave-pinning, 0.798–1.000 in Goryachev and 0.921–0.999 in Holmes.
- The complete finite maps contain 81 Wave-pinning and 82 Holmes distinct-old-plus-cue outcomes, while Goryachev has none in this finite grid. Its native C40/C80 antipodal histories instead distinguish replacement from old-region return/retention.
- Local patterns and global activity are complementary. Opposing active sites can cancel the first moment without loss of local activity; conversely, a strong moment does not identify a unique site or distinguish a merger from a shifted peak.

Proposed measurements and experimental follow-up

Vary stimulation location at fixed measured size and initial reporter organization; record physical center separation, site extent, edge gaps, both available boundary routes and input overlap.

Follow incumbent and challenger activity, extinction/merger, distinct-site lifetimes and local loss/rebound alongside the whole-boundary reporter-polarity vector, magnitude and direction. A fixed-region loss can be redistribution, so retain tracked sites and neighboring signal.

Figure 3. Challenger placement changes sustained competition and global activity polarity on supported rings · continued

For irregular cell boundaries, declare registration, physical arclength weights and vector embedding. Pair activity with local boundary extension/retraction to test protrusion correspondence without assuming a velocity law.

Scope, limitations and pending validation

Moving a cue in a fixed basal changes both routes and local cortical context; it does not isolate transport distance. Cross-size comparisons also change material, architecture and angular input.

Common local-recovery responses and finite outcomes are different measurements, with no demonstrated causal allocation of reversal time. Near-zero vector magnitude must not be assigned a meaningful heading.

Native amplitudes and times are not cross-calibrated, and ring circumference is not silently equated to cell diameter. Finite deterministic vectors do not estimate stochastic polarity persistence or migration.

Model definitions and readouts

$$M_k(t) = \int_0^C u(s,t) e^{2\pi i k (s - s_0)/C} \, \mathrm{d}s \tag{3}$$

Circular activity moments retain physical boundary weighting. The same fixed origin is used throughout a history; uniform background cancels from the first moment.

$$P(t) = \frac{M_1(t)}{M_1(0)}, \qquad R(t) = |P(t)|, \qquad q_k(t) = \frac{|M_k(t)|}{M_0(t)} \tag{4}$$

Relative activity magnitude and mass-normalized harmonic magnitudes remain distinct. The complex division aligns the initial first moment to (1,0).

Sources and reproducibility

Frozen initial checkpoint intake
Actual supported-ring report
Common-assay ring recovery results
Native interval/ring display profiles
Complete conditional ring recovery
Recovery and vector methods
Actual-result scientific critique
Assembly provenance

Figure 4. Noisy-ring activity memory and physical motion · image 1 of 4

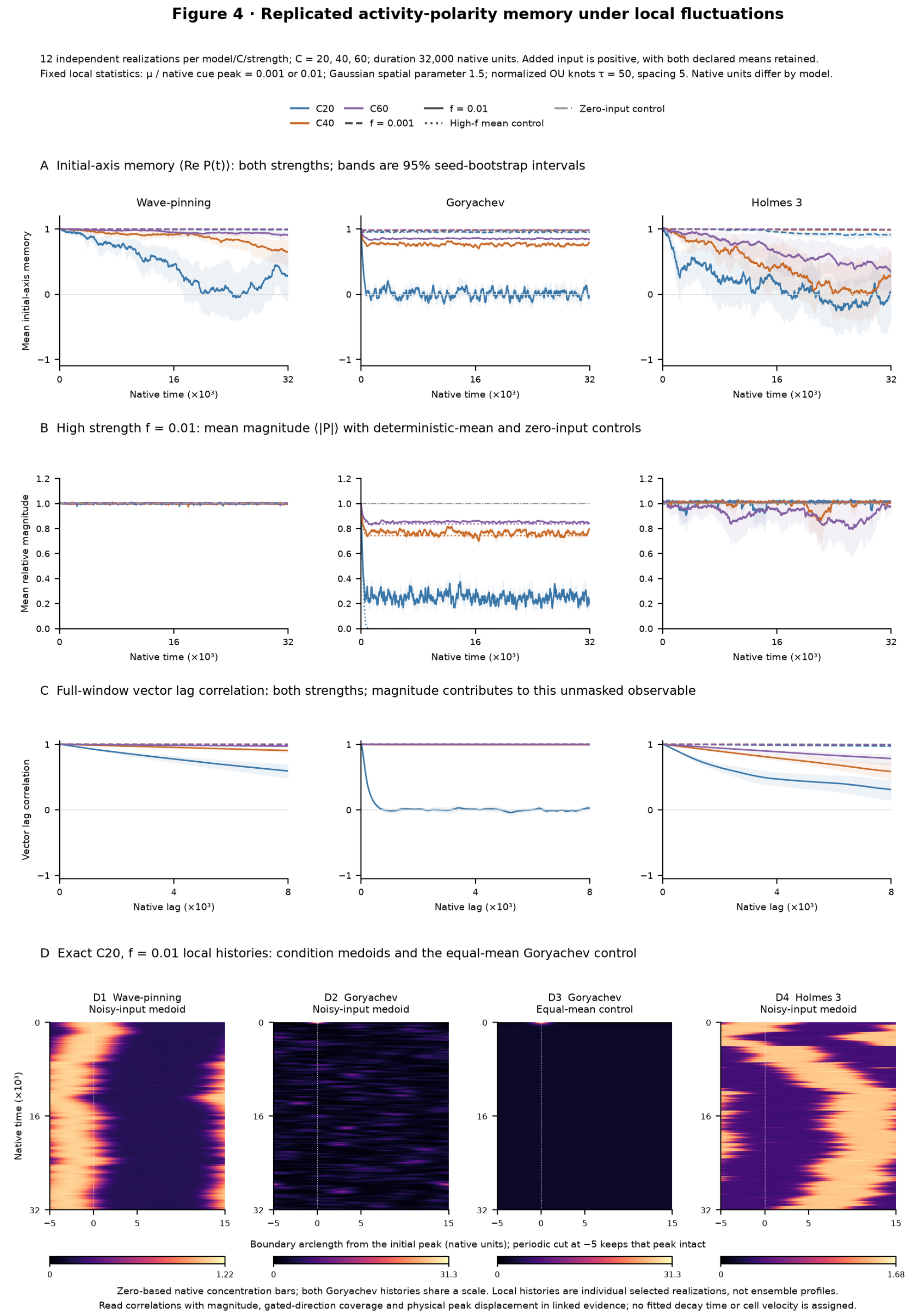
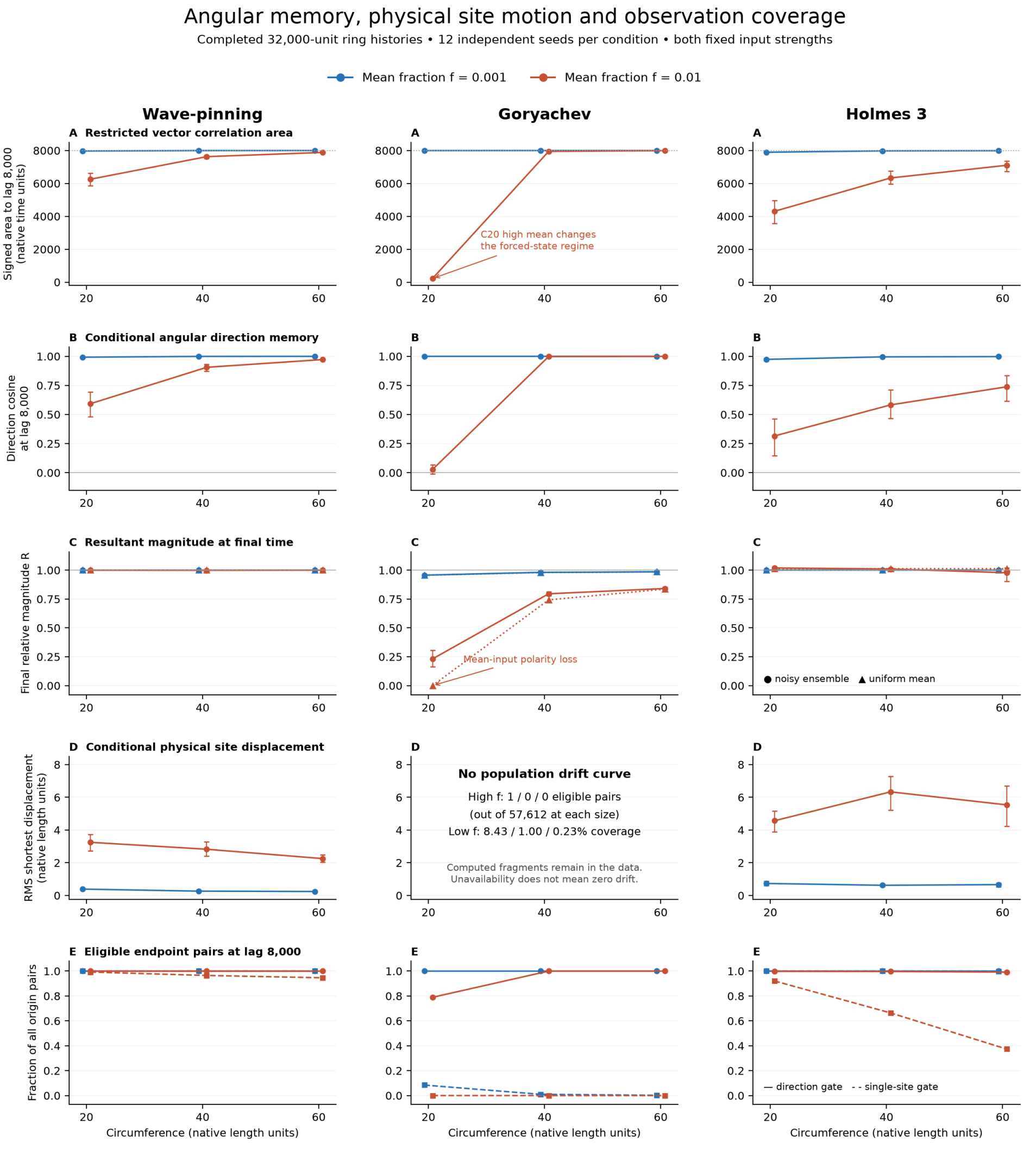


Figure 4 memory/time-course sheet: original numerical results with the working title renumbered.

Figure 4. Noisy-ring activity memory and physical motion · image 2 of 4

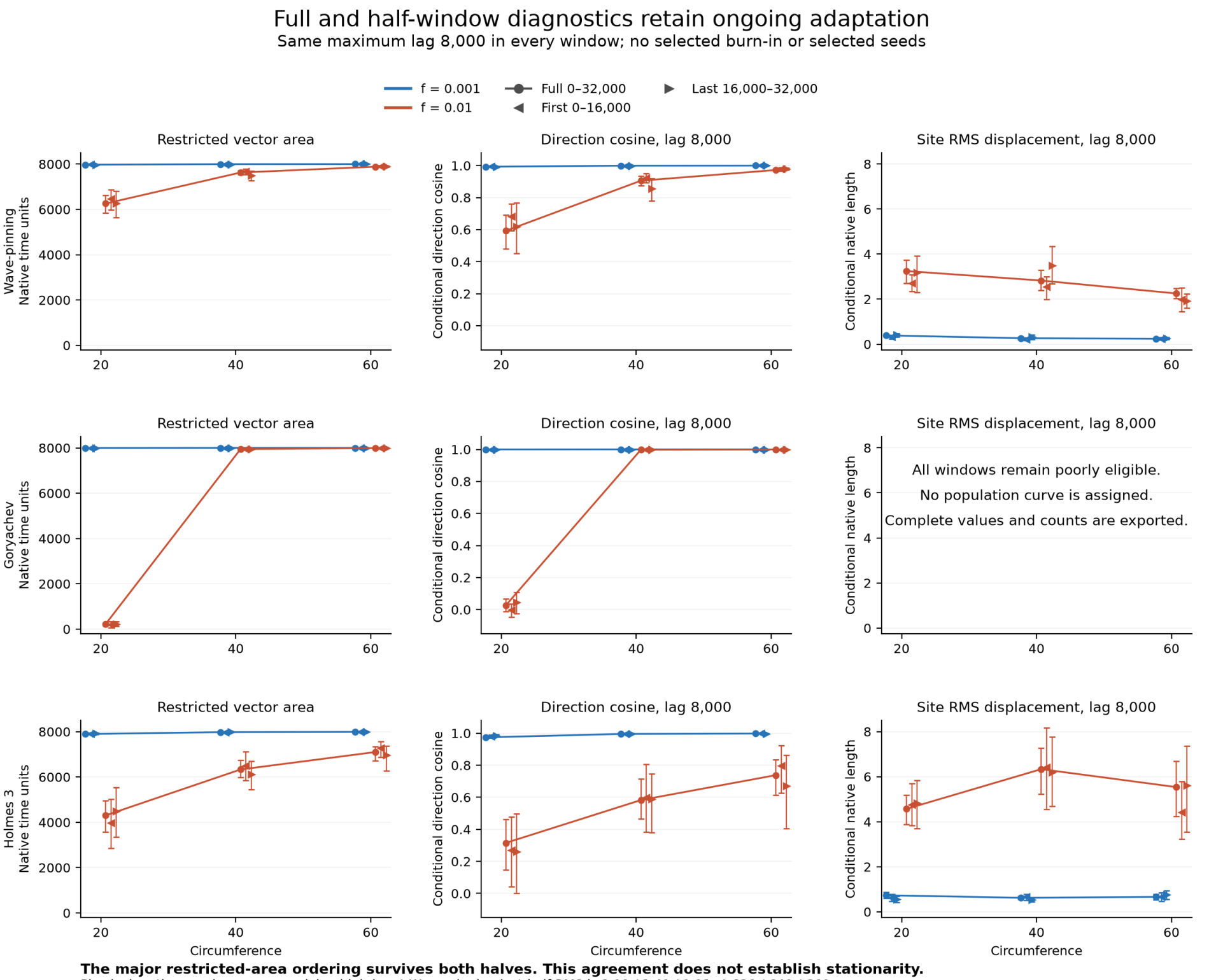


Angular geometry: Δθ = 2π Δs / C. Equal physical displacement subtends a smaller angle on a larger boundary.
Wave-pinning: high-input physical displacement is smaller at C60 than C20. Holmes: higher angular memory coexists with nonmonotone physical displacement and decreasing single-site coverage. Goryachev requires a forced-state qualification.

Bars: source 95% whole-seed bootstrap intervals. Signed finite-window areas are not fitted or extrapolated persistence times.
Site endpoints must pass the complete local-profile and R ≥ 0.1 gate; continued site identity is unproven. All windows are retained in the companion sheet.

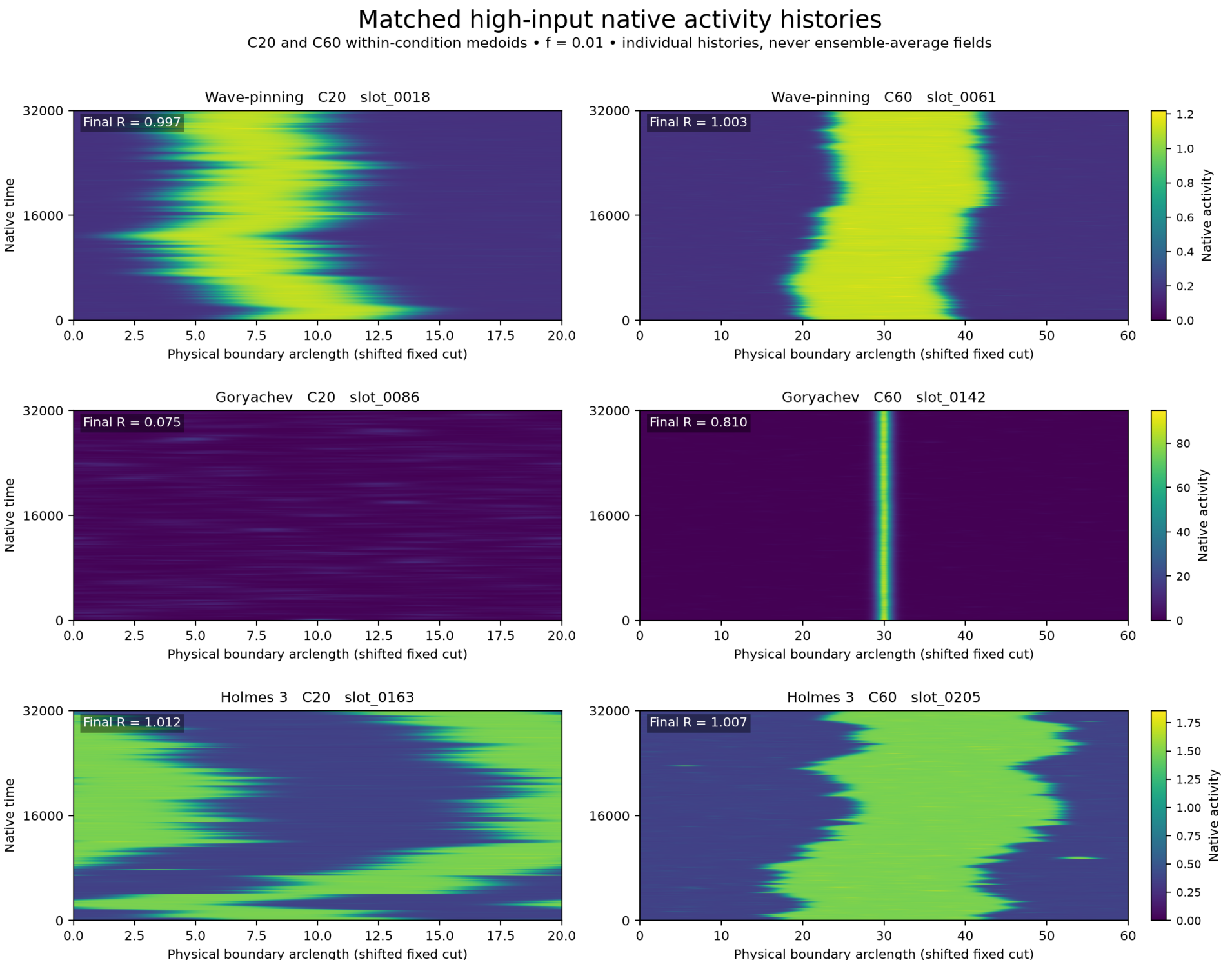
Continued Figure 4: finite memory, angular geometry, native magnitude and observation coverage; both input strengths.

Figure 4. Noisy-ring activity memory and physical motion · image 3 of 4



Supporting/continued sheet: full/first/last windows with the same lag horizon; no selected burn-in.

Figure 4. Noisy-ring activity memory and physical motion · image 4 of 4



Supporting/continued sheet: exact high-input C20/C60 within-condition medoid histories, matched zero-based scale within each model.

Figure 4. Activity memory, physical site motion and forced-state qualification on noisy rings

A direct stochastic polarity-memory test, separated from deterministic cue resistance and from conditional OU velocity-memory estimates in the migration evidence.

Detailed legend

Memory/time-course sheet: A, Ensemble mean initial-axis memory, the real part of the first activity moment divided by its own initial complex value, for Wave-pinning/Goryachev/Holmes rings at circumferences 20/40/60. Each model/size/strength has 12 independent declared seeds, followed for 32,000 native units. Colors denote circumference; dashed/solid curves denote mean input fractions 0.001/0.01 of the model's native cue peak. Bands are 95% seed-bootstrap intervals. Time origins within a trace are not independent experimental replicates.

Memory/time-course sheet: The positive native activation input is lognormal, with unit-variance Gaussian field, physical spatial correlation parameter 1.5, temporal OU-knot correlation time 50 and knot spacing 5. Linear interpolation is divided by its analytic marginal standard deviation. The law is a normalized interpolated stochastic process, not an exact continuous-time OU path. Mean fraction and local spatial/temporal statistics are fixed across sizes within each model; amplitudes and times are not cross-model biochemical calibrations. No realization is demeaned, clipped, circumference-normalized or renormalized around its own realized mean. Larger rings contain more fluctuating boundary, so total input and whole-ring angular forcing modes can differ.

Memory/time-course sheet: B, Mean relative moment magnitude at the higher strength, with the corresponding deterministic uniform-mean and zero-input controls. Both strengths and all controls remain in the linked evidence. A uniform input rate can alter activity moments through the patterned inactive substrate and can change the forced state even when its own input harmonic is zero. A diminished first moment must be read with raw local activity, total amount and higher harmonics.

Memory/time-course sheet: C, Full-window uncentered vector lag correlation through common lag 8000, retaining magnitude fluctuations and weak resultants. Equal-seed averages of within-seed time-origin products are normalized by both endpoint energies. Replicate bootstrap recomputes that ratio. Full-, first-half and last-half windows, endpoint energies and magnitude-gated direction coverage remain linked. The signed restricted integral through 8000 is a finite-window descriptor; no forced exponential fit or unobserved persistence time is assigned.

Memory/time-course sheet: D, Exact C20, high-strength local histories for the within-condition medoids in Wave-pinning, Goryachev and Holmes, plus the Goryachev deterministic equal-mean control. Medoids minimize the declared standardized distance to within-condition descriptor medians among 12 seeds, using final initial-axis memory, minimum and mean magnitude, and multiregion fraction; ties follow the case identifier. They are individual selected histories, not ensemble-average fields. The periodic cut

Figure 4. Activity memory, physical site motion and forced-state qualification on noisy rings · continued

keeps the initial peak intact. Native color scales start at zero; the two Goryachev histories share a scale.

Memory/time-course sheet: Independent audits check native source/input/output hashes, exact basal starts, complete scalar/profile time grids, moments, site descriptors, pool behavior and prescribed coverage. A separately reviewed numerical amendment evaluates the same frozen correlation estimator using positive suffix energies and direct products when tiny-tail FFT calculations are unresolved; the original frozen source is preserved. Undefined zero-energy correlations stay undefined. Weak endpoint RMS is flagged and is not promoted to a heading. The validation-only flat-centroid amendment preserves strict agreement on all nonflat saved profiles (maximum error zero across the completed audit); all 697 flat saved frames carry separately checked cache labels and are ineligible for physical site claims.

Diagnostic/continued sheets: A–E, Completed noisy-ring histories from three native periodic chemical models at circumferences C20/C40/C60. Every model/size/strength contains 12 independent realizations, each observed for 32,000 native time units. Blue/red denote the two prospectively frozen mean input fractions f=0.001/0.01. The driving law is positive lognormal activation with fixed local spatial covariance parameter 1.5 and analytically variance-normalized interpolation of OU knots (correlation time 50; spacing 5). No per-realization mean removal or circumference renormalization is applied. Local input statistics are fixed within model, while total input and angular forcing modes vary with circumference. Native amplitudes and times are not cross-model biological calibrations.

Diagnostic/continued sheets: A, Actual signed full-window vector-correlation area from lag 0 through 8000 on the 5-unit grid. The uncentered vector ratio retains magnitude fluctuations and weak resultants, subtracts no average vector and normalizes by both endpoint energies. The dotted 8000 line is the maximum area for correlation identically one; proximity marks unresolved decay. Intervals are stored 95% whole-seed bootstrap percentile bounds on the integrated recomputed correlation ratio. Sixteen of 18 conditions do not cross 1/e within this lag horizon. No fitted exponential time, extrapolated persistence time or cell motility time is assigned.

Diagnostic/continued sheets: B, Direction-only cosine at the same lag 8000 and over the same full time-origin window. Both endpoints require relative moment magnitude R>=0.1. Bars are stored pointwise 95% whole-seed bootstrap intervals. The eligible fraction of all possible endpoint pairs is shown in E. Conditional angular memory and raw vector area answer distinct questions.

Diagnostic/continued sheets: C, Mean relative first-moment magnitude at the final time 32000 (circles), with the matched deterministic uniform-mean input controls (dotted triangles). This final-time magnitude is a forced-state diagnostic, not the lag-correlation denominator. Initial R = 1; R can exceed one. In Goryachev high-input C20 the

Figure 4. Activity memory, physical site motion and forced-state qualification on noisy rings · continued

uniform-mean control ends with essentially zero resultant; larger rings retain substantial polarity. Its small vector area cannot be attributed solely to noise-driven wandering around an otherwise unchanged polarized state.

Diagnostic/continued sheets: D, Square root of the conditional mean squared shortest physical dominant-region center displacement at lag 8000; confidence bounds are monotone square roots of the source squared-displacement intervals. Both endpoints must have exactly one current half-contrast connected region, at most one peak with native prominence>=0.05, contrast>=0.05, and R>=0.1. Site descriptors are recorded every 5 native units and were independently validated at every saved 40-unit primary profile in the source audit. Endpoints do not establish uninterrupted site identity. All actual Goryachev values remain in the exported tables; the population curve is withheld at both strengths because eligibility is sparse. High input supplies only 1/0/0 eligible pair at C20/C40/C60 among 57,612 possible pairs per condition, and low-input coverage is 8.43/1.00/0.23%. The isolated one-pair result is not assigned a population precision interval or represented as zero drift.

Diagnostic/continued sheets: E, Fraction of all available lag 8000 endpoint pairs satisfying the direction gate (solid circles; source seed-bootstrap intervals) or the complete physical-site gate (dashed squares; exact counts exported). At high input, Holmes physical-site coverage decreases 92.1%→66.4%→37.3%, although direction coverage remains above 99%. The declared absolute-prominence gate is deliberately conservative: small secondary peaks may fail it despite a dominant incumbent. Eligibility is an observation restriction, not a functional competitor count or an instability metric.

Diagnostic/continued sheets: Companion adaptation sheet, full [0, 32000], first [0, 16000] and last [16000, 32000] windows retain identical maximum lag 8000 and inclusive endpoints. Symbols distinguish windows; colors retain strengths. The source estimator and all 12 seeds are unchanged, with no chosen burn-in. All windows and coverage counts remain exported. Agreement of major area ordering does not establish stationarity; physical displacement is less uniformly ordered.

Diagnostic/continued sheets: Companion native sheet, exact high-input C20/C60 within-condition medoids for all three models. Medoids minimize the sum of absolute deviations from within-condition descriptor medians, scaled by IQR (unit fallback when IQR $\le 10^{-12}$), for final initial-axis memory, minimum R, time-mean R and multisite-frame fraction. Ties use case identifier. Each model uses one shared zero-based native activity scale across sizes; physical arclength axes retain circumference. The static plotting cut is shifted so the initial maximum lies at mid-boundary. The exact full-resolution arrays, shift indices, descriptor distances, native source hashes and trace arrays are included. These individual histories illustrate the quantitative population evidence without supplying independent uncertainty.

Figure 4. Activity memory, physical site motion and forced-state qualification on noisy rings · continued

Narrative points and conclusions

- At high input, the measured full-window restricted vector areas rise C20 → C40 → C60 in Wave-pinning 6257 → 7627 → 7885 and Holmes 4304 → 6332 → 7103 native time units. The direction-only comparisons retain the size ordering, while large-ring direction remains measurable. This supplies a direct, declared-noise activity-memory observation alongside deterministic replacement experiments; it does not close the causal migration-persistence hypothesis.
- The physical interpretation differs by model. High-input Wave-pinning full-window physical RMS is 3.25 [2.71,3.74],2.83 [2.40,3.29],2.25 [2.03,2.48] at C20/C40/C60, so the larger/smaller endpoint contrast includes reduced physical displacement under the declared gate. Holmes instead gives 4.57 [3.88,5.17],6.33 [5.22,7.28],5.54 [4.25,6.70] with decreasing coverage: angular memory alone cannot establish slower physical-site motion. The geometric identity $\Delta\theta = 2\pi\Delta s/C$ explains why equal physical displacement is less angular change on a larger ring, without assigning the observed effect to geometry causally.
- Goryachev high-input C20 is a distinct forced-state regime, not a clean noise-specific reorientation comparison around the same maintained polarity. Conservative site eligibility is sparse even at low input and effectively absent at high input. This leaves physical comparison unavailable rather than proving stationary local activity; the raw histories can still reveal a strong dominant incumbent or locally structured activity.
- The lower input mostly leaves decay unresolved over lag 8000. It is retained as an informative weak-response regime; apparent areas near 8000 are bounded observation descriptors rather than arbitrarily long estimated memory times.
- Full/first/last restricted areas retain the major size ordering, but physical motion needs a bounded endpoint statement. Wave-pinning high-input last-half RMS is 3.19/3.49/1.93 at C20/C40/C60, so a universal monotone size rule is unsupported. Holmes C60 high-input direction changes 0.798 first-half → 0.670 last-half. These windows are shown rather than silently selecting a favorable stationary segment.

Proposed measurements and experimental follow-up

In independent cell recordings, jointly measure whole-boundary activity vectors, their magnitudes, total reporter activity and local region/peak histories over a declared observation horizon and cadence. Report weak-magnitude fractions, pair coverage and finite-window restricted correlations; compare early and late windows before interpreting a stationary time.

Register physical boundary arclength and track local activity-site displacement separately from angular reorientation. Retain multisite, low-contrast and weak-resultant intervals in

Figure 4. Activity memory, physical site motion and forced-state qualification on noisy rings · continued

the coverage denominator and report unavailable single-site comparisons explicitly. Calibrated signal is needed to interpret integrated fluorescence as molecular amount.

Measure local protrusion or extension/retraction alongside reporter activity to test whether persistent or renewed activity sites correspond to protrusive behaviour. Then associate these independently measured quantities with matched cell trajectories; do not impose a velocity law on the polarity model.

Relate the noisy-ring distinction to the primary cue comparison: quantify early challenger activation, incumbent loss/rebound, replacement/return/coexistence and local site lifetime after controlled cues, alongside spontaneous activity fluctuations. A local recovery response, finite cue resistance and spontaneous activity memory remain separate observables.

Scope, limitations and pending validation

The input process is a declared perturbation ensemble, not a measured cell-noise law. Changed circumference also changes total forcing and angular forcing modes, material, basal architecture and the geometric embedding, so no causal allocation among these changes is demonstrated.

The restricted correlation is uncentered and may retain a nonzero average axis or adaptation. All windows remain finite; absence of a 1/e crossing gives no resolved exponential time. Confidence intervals quantify independent-seed variation at the fixed native model/input condition, not model discrepancy or biological uncertainty.

Site displacement is shortest endpoint distance modulo C, not unwrapped travel, diffusion, cell movement or proof of uninterrupted site ancestry. Gate failure can be caused by small secondary peaks under the conservative prominence rule; unavailable comparisons are never zeros.

Activity-vector memory, conditional physical site displacement and conditional OU velocity memory remain distinct. No cell-velocity law or calibrated migration-persistence time follows from these rings.

Model definitions and readouts

$$I(s,t) = fA_m \exp\left[Z(s,t) - \tfrac{1}{2}\right] \tag{5}$$

The ensemble local input mean is f A_m; f is 0.001 or 0.01 and A_m is the native cue peak 0.1, 6.4 or 1 for Wave-pinning, Goryachev or Holmes.

Figure 4. Activity memory, physical site motion and forced-state qualification on noisy rings · continued

$$C_W(\lambda) = \frac{\langle \mathrm{Re}[P(t+\lambda)P(t)^*]\rangle_W}{\sqrt{\langle |P(t)|^2\rangle_W \langle |P(t+\lambda)|^2\rangle_W}} \tag{6}$$

Brackets denote equal-realization averages of within-realization sampled origins whose two endpoints lie in W. No mean vector is subtracted.

$$T_W^{\mathrm{restricted}} = \int_0^{8000} C_W(\lambda)\,\mathrm{d}\lambda \tag{7}$$

The raw signed correlation is integrated on the prescribed 5-unit lag grid. This descriptor retains negative values and is not extrapolated beyond the measured horizon.

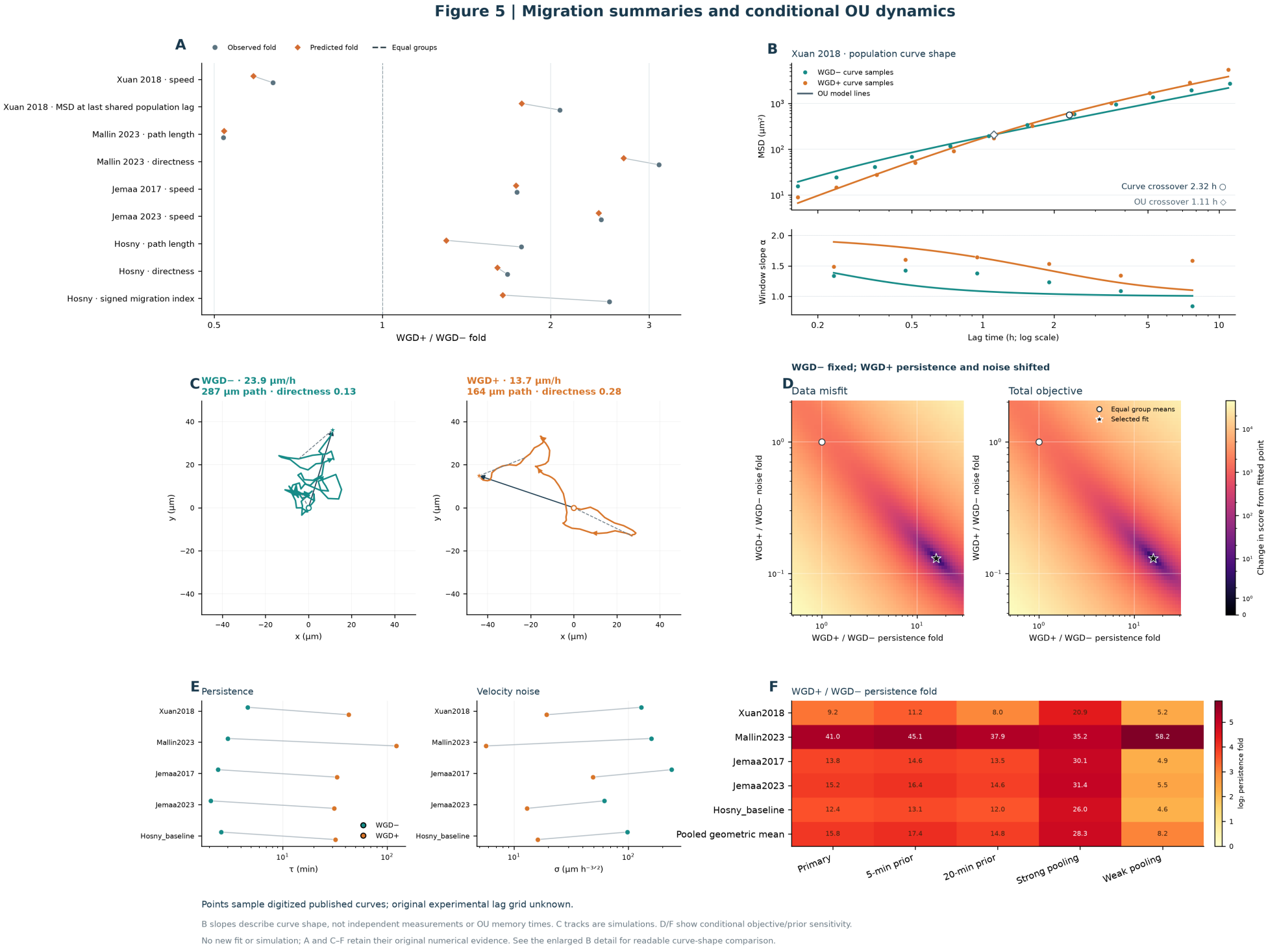
$$\Delta\theta = \frac{2\pi\Delta s}{C} \tag{8}$$

For the same boundary arclength displacement, angular change decreases with circumference. The measured shortest endpoint displacement is conditional on the stated single-site gate; it is not an unwrapped trajectory or a cell displacement.

Sources and reproducibility

Frozen initial checkpoint intake
Replicated noisy-ring actual results
Frozen noise and statistical definitions
Complete small seed summary
Restricted integrals and weak-window statuses
Complete numerical and source audit
Validation-only flat-centroid correction
Pilot regimes and numerical checks
Assembly provenance
Source/native reconciliation and exact pair counts
All plotted source rows and hashes
Compact native histories and selection
Complete compact lag exports and provenance

Figure 5. Migration observations and conditional OU interpretation · image 1 of 2



Reassembled Figure 5: original A and C–F numerical evidence, revised MSD/curve-shape panel B.

Figure 5. Migration observations and conditional OU interpretation · image 2 of 2

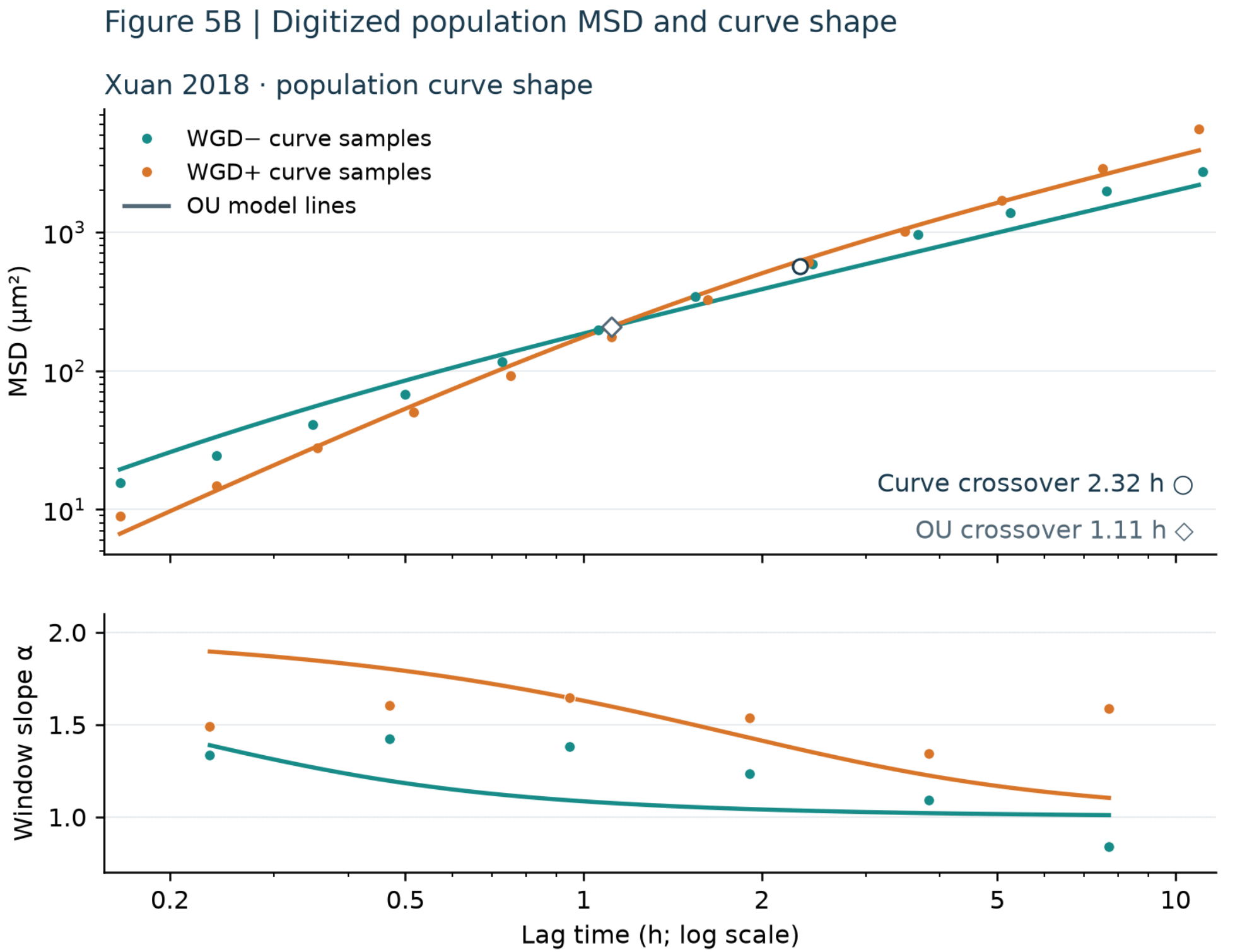


Figure 5B continued: enlarged detail with explicit raster-sample caveat.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Revised display, exact preserved inputs and direct-user feedback
Migration display numerical and preservation checks

Figure 6. RESERVED: new WGD motility results



Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 6. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 6. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

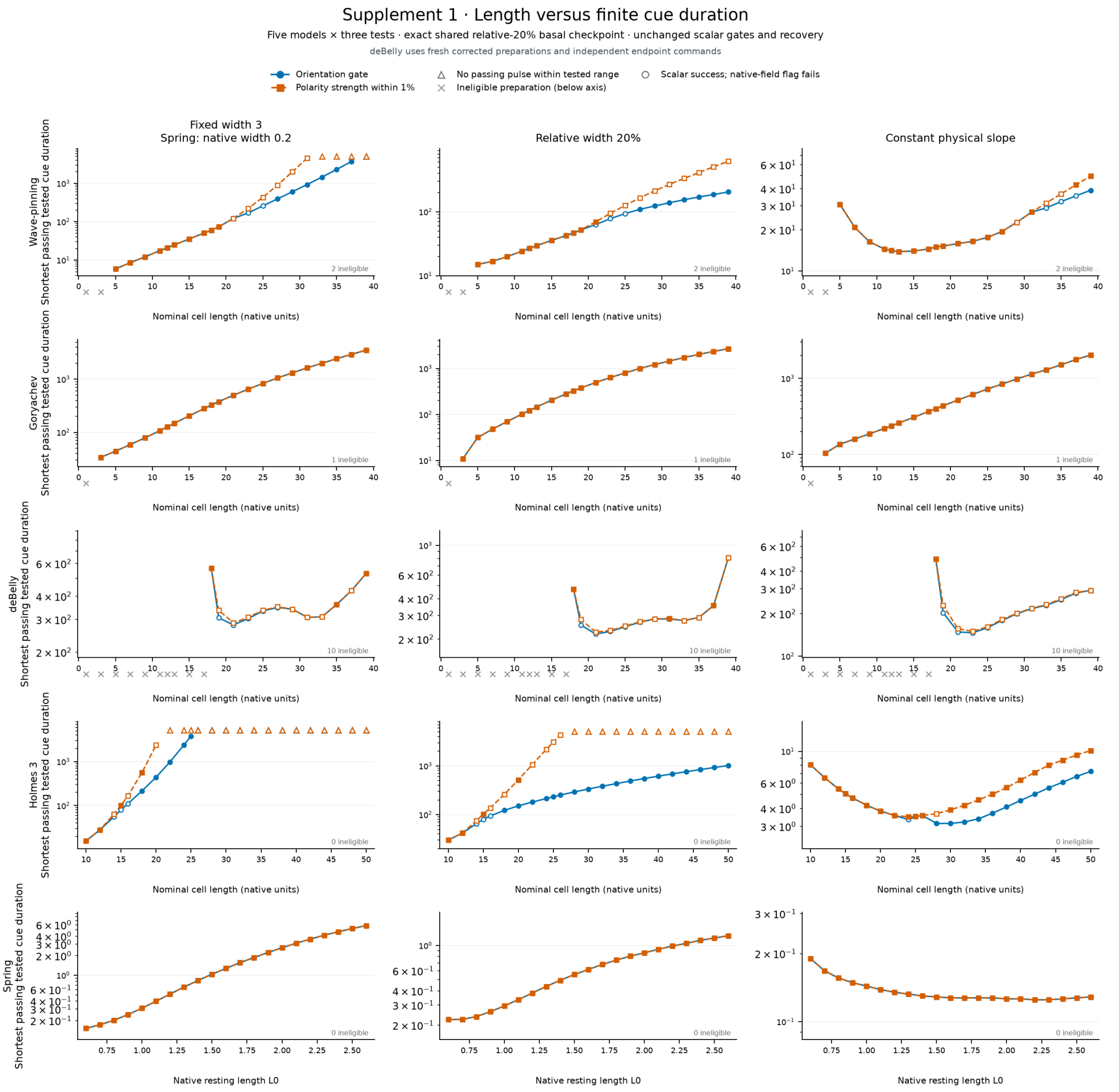
Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

S1. Full cue-duration response curves



Both patches use a constant peak across all lengths; Goryachev K=64, peak=6.4. Gradients retain one constant physical slope per model.
Each positive duration restarts the original checkpoint. Full strength adds the 1% reversed initial-polarity band to the orientation gate.
Open markers retain failing field-tail flags. Censoring marks explicit tested limits; missing endpoints break lines. Native times and scales differ.
Data generated 2026-10-08T19:26:33.200718+00:00; producer finished. Every imported and fresh trial passed independent native-array audit.

Canonical corrected five-model duration curves; source working number Supplement 1.

S1. Full cue-duration response curves

Essential curve context for the main comparison heatmap.

Detailed legend

Five model rows and three finite-cue columns retain the complete sampled length dependence. Coloured curves distinguish orientation from strength replacement. Open success markers retain scalar successes that fail the separate native-field tail flag; triangles indicate no passing pulse within the tested range; crosses indicate ineligible preparations. Missing values are not zero response times. Each duration excludes preparation and the subsequent 10,000-unit basal recovery. Spring uses native length and time. See Figure 1 for exact gates and stimulus conventions.

Narrative points and conclusions

- Corrected deBelly is nonmonotone across all three geometries. The positive rank coefficients summarize association rather than its full curve shape.
- Constant-slope Spring shows the opposite association to its patch responses. Holmes orientation under that cue varies with little monotone association. Censoring and native-field flags restrict interpretation even where rank correlation is strong.

Sources and reproducibility

Current figure legend and gate definitions

Supplementary material and supporting analyses

S2. Numerical qualification and preparation provenance

Exact common starting states, duration-boundary audits, corrected deBelly endpoint controls and selected mesh diagnostics; source histories remain distinguishable from compact committed summaries.

polarity1d/manuscript/THREE_TESTS.md

S3. Mechanism and stochastic sensitivity

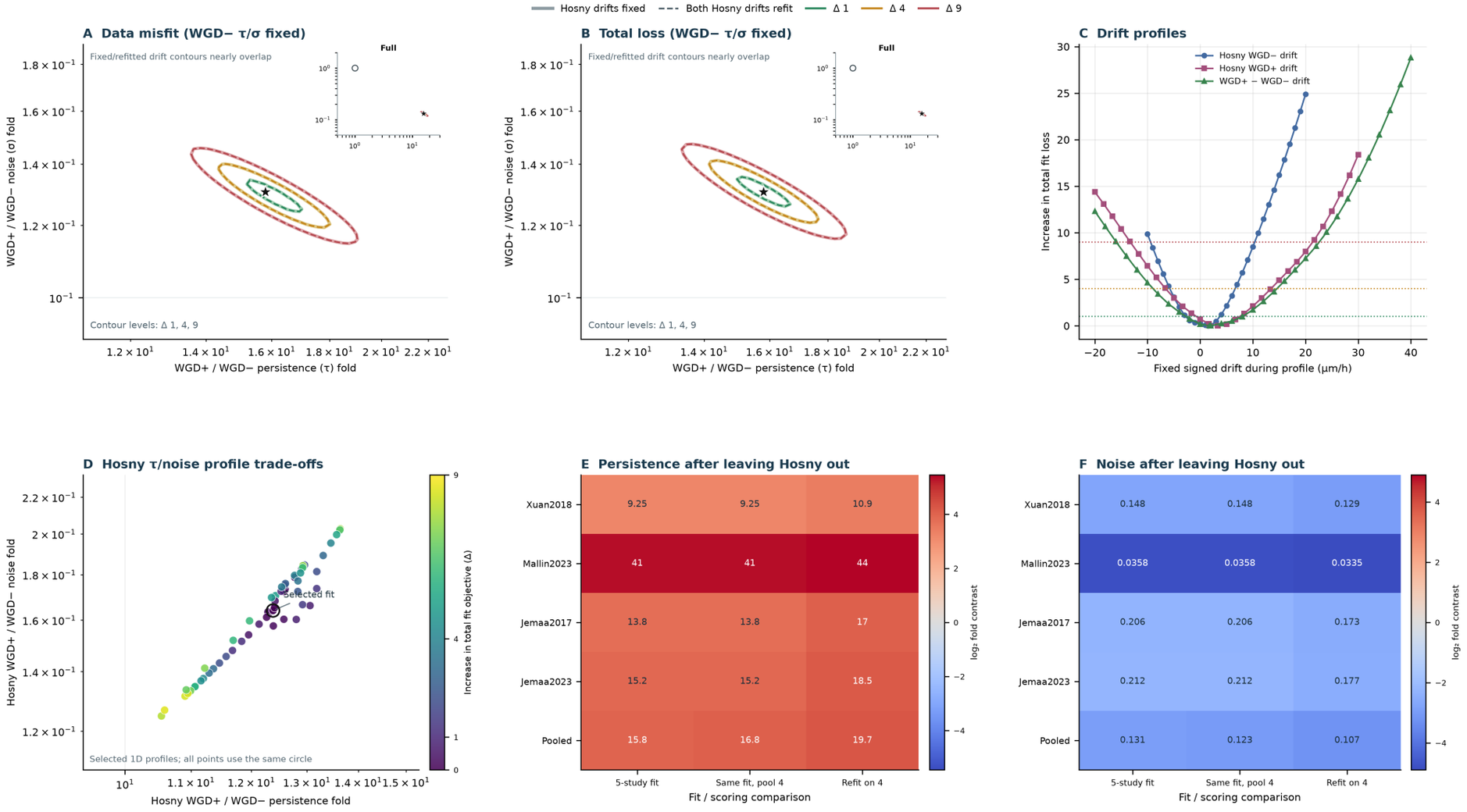
Complete gain/recovery grids, clock qualification, finite-cue definitions, outcome tables, site/vector diagnostics and numerical/noise controls accompany the mechanism checkpoint. Revised Figure 4 also displays both strengths, complete half-window diagnostics, physical-site eligibility and matched C20/C60 native examples; compact source tables support portable rerendering.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/VISION_CHECKPOINT_REQUIREMENTS.md

S4. Migration-model adequacy and Hosny sensitivity

How much does the Hosny drift assumption change the fitted contrast?

Primary Gaussian OU hierarchy · profile panels reoptimize remaining parameters · Δ values are descriptive fit sensitivity



Objective is arbitrarily weighted and has no calibrated statistical or biological units; it is not a likelihood. Profile ranges are unions of sampled 1D fits, not joint uncertainty regions. Hosny contributes medians to fitting.

Existing Hosny sensitivity supplement from the refined OU analysis.

S4. Migration-model adequacy and Hosny sensitivity

Supporting conditional sensitivity analysis; it is not an independent biological replication.

Detailed legend

The retained supplement explores signed-drift and leave-Hosny-out sensitivity within the declared OU hierarchy. Its fitted changes and descriptive-score profiles are conditional on the model and supplied summaries, not confidence intervals. Data-only identifiability, Xuan population-curve residuals and prior/pooling assumptions remain part of the adequacy assessment. Representative movie diagnostics are separate: acquisition timing is uncertain and nuclear appearance does not establish ploidy.

Sources and reproducibility

Original Hosny supplement caption

Sensitivity methods and results

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope